

1. EC2 Purchasing Options (EC2 Purchasing Cost)

AWS provides different ways to pay for EC2 instances depending on your usage.

A. On-Demand Instances

- Pay only for the time you use the instance.
- No long-term commitment.
- Best for short-term or unpredictable workloads.

Advantages

- Flexible
 - No upfront payment
 - Easy to start and stop
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B. Reserved Instances (RI)

- Reserve an instance for **1 or 3 years**.
- Lower cost than On-Demand.
- Best for applications with predictable usage.

Advantages

- Up to **72% cost savings**.
 - Suitable for long-running applications.
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C. Spot Instances

- Use AWS's unused EC2 capacity at a very low price.
- AWS can terminate the instance if capacity is needed.

Advantages

- Up to **90% cheaper** than On-Demand.
 - Best for testing, batch processing, and data analysis.
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D. Savings Plans

- Commit to a fixed amount of usage for **1 or 3 years**.
- Flexible across different EC2 instance types.

Advantages

- Lower cost than On-Demand.
 - More flexible than Reserved Instances.
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E. Dedicated Hosts

- A physical server dedicated to a single customer.
- Used when organizations require dedicated hardware.

Advantages

- High security.
 - Meets licensing and compliance requirements.
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2. Amazon S3 Bucket Creation

What is an S3 Bucket?

An **Amazon S3 Bucket** is a container used to store objects (files, images, videos, backups, documents) in AWS.

Steps to Create an S3 Bucket

1. Log in to the **AWS Management Console**.
2. Open **Amazon S3**.
3. Click **Create Bucket**.
4. Enter a **Bucket Name** (must be globally unique).
5. Select an **AWS Region**.
6. Configure options (Versioning, Encryption, etc.).
7. Set **Bucket Permissions**.

8. Click **Create Bucket**.

Features of S3 Bucket

- Unlimited object storage.
- High durability (99.999999999% durability).
- Scalable.
- Secure with IAM policies and bucket policies.
- Supports versioning and lifecycle management.

3. IAM User Creation

What is an IAM User?

An **IAM (Identity and Access Management) User** is an individual user who can securely access AWS resources.

Steps to Create an IAM User

1. Open the **AWS Console**.
2. Go to **IAM**.
3. Select **Users**.
4. Click **Create User**.
5. Enter a username.
6. Choose access type (Console access and/or Programmatic access).
7. Assign permissions (directly or through a group).
8. Review and click **Create User**.

Benefits

- Secure access to AWS.
- Separate login credentials for each user.

- Fine-grained permission control.
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4. IAM Groups

Definition

An **IAM Group** is a collection of IAM users. Permissions assigned to the group are automatically inherited by all users in that group.

Example

Developers Group

└─ User1

└─ User2

└─ User3

If the **Developers Group** has EC2 access, all members receive EC2 access.

Benefits

- Easy permission management.
 - Reduces administrative work.
 - Consistent access control.
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5. IAM Roles

Definition

An **IAM Role** is an AWS identity that provides temporary permissions to AWS services or users without requiring long-term credentials.

Example

An EC2 instance needs permission to access an S3 bucket.

Instead of storing AWS access keys inside the instance, attach an **IAM Role** to the EC2 instance.

Uses

- EC2 accessing S3.
 - Lambda accessing DynamoDB.
 - Cross-account access.
 - Temporary access for applications.
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Benefits

- Temporary credentials.
 - Improved security.
 - No need to store access keys.
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6. IAM Policies

Definition

An **IAM Policy** is a JSON document that defines **what actions are allowed or denied** on AWS resources.

A policy specifies:

- **Who** can access.
 - **What actions** they can perform.
 - **Which resources** they can access.
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Types of IAM Policies

A. Managed Policies

Policies created and managed by AWS or by the customer.

Types

1. AWS Managed Policy

- Created and maintained by AWS.
- Example:

- AmazonS3ReadOnlyAccess
- AmazonEC2FullAccess

2. Customer Managed Policy

- Created and managed by your organization.
- Can be reused across multiple users, groups, and roles.

B. Inline Policies

- Directly attached to a single user, group, or role.
- Cannot be shared.
- Deleted automatically if the identity is deleted.

Difference Between Managed Policy and Inline Policy

Managed Policy	Inline Policy
Separate policy object	Embedded in one user/group/role
Can be attached to multiple identities	Attached to only one identity
Reusable	Not reusable
Easier to manage	Used for specific permissions

Difference Between IAM User, Group, Role, and Policy

IAM User	IAM Group	IAM Role	IAM Policy
Individual AWS user	Collection of users	Temporary AWS identity	Permission document
Has permanent credentials	Shares permissions	Temporary credentials	Defines Allow/Deny permissions
Login to AWS	Simplifies permission management	Used by AWS services and users	Attached to users, groups, or roles

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- **EC2 Purchasing Options:** Different pricing models for EC2 such as **On-Demand, Reserved, Spot, Savings Plans, and Dedicated Hosts.**
 - **S3 Bucket:** A container in Amazon S3 used to store and organize objects (files).
 - **IAM User:** An individual identity that can securely access AWS resources.
 - **IAM Group:** A collection of IAM users with shared permissions.
 - **IAM Role:** A temporary AWS identity that provides permissions to users or AWS services.
 - **IAM Policy:** A JSON document that defines permissions (Allow or Deny) for AWS resources.
 - **Types of IAM Policies:** **AWS Managed Policies, Customer Managed Policies, and Inline Policies.**

I created this web page using s3 bucket in aws

